

Amlogic

NN Model Design Guide

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Revision History

Issue 1.0 (2022-10-12)

This is the initial release.

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1. Introduction

This document is intended to provide some design guidance for customers who have the ability to design the model structure and train the model by themselves, and guide customers to design a model with better acceleration effect on Amlogic NN.

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2. Model Acceleration Rule

1. Depth convolution is recommended

Amlogic NN Hardware has both DW Engine and Conv Engine, regular 2D convolutions and Depthwise convolutions can run on Hardware at the same time. Amlogic NN Hardware is optimized to run regular Conv2D & DepthWiseConv2D in a pipeline style. Customers can get the best performance if they run an AI model which is similar to Mobilenet. Recommend to use PW and DW convolution.

2. Padding has no software overhead

Convolution padding is supported by NN hardware with no software overhead. The padding attribute can be specified individually for all four sides [Top, Bottom, Left, Right].

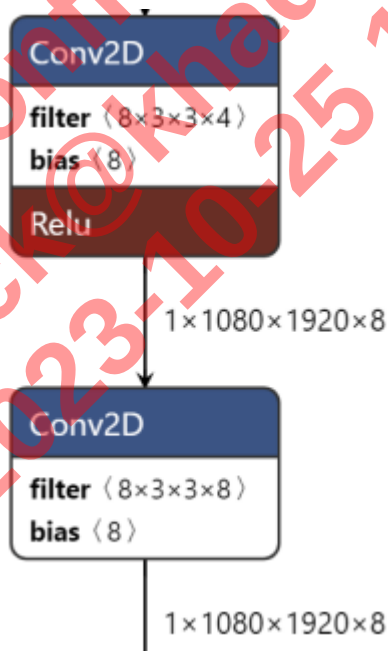
The maximum pad size is 16.

3. Use pooling instead of stride

Whenever possible for strides not of 2, design the convolution kernels without stride and use pooling operations to achieve feature map shrinking.

4. Conv OP layer which with output more than 16 channels

ADLA is designed to maximum parallel execution and minimize the power and cost. It requires a OP layer which will output more than 16 channels to fully utilize the horsepower of ADLA. For example, the following figure shows a customer's AI model contains 8 output channels only.



5. Design Suggestions for Other Operators level

a . Free Op

The following OP consumes neglectable execution time:

Activation function & Quantization & DeQuantization

b . Convolution

Horizontal or vertical stride can be set independently, But the best performance can be gain if stride size is no more than 2

c . Resize tensor

Recommend resize ratio is power of 2

d .Other Op

It is recommended to use operators running on NN Hardware to design models as much as possible

6. Take advantage of Hardware' s Sparse Matrix Support

Modern Neural-Networks are known to be over parameterized and have much redundancy in their design. Pruning a network to be sparse has been proven to reduce computation overhead while maintaining accuracy. AMLNN hardware is designed to support sparse matrix operations efficiently by skipping computations and memory fetches on zero values. The sparsity level can be fine grain down to individual weights. Designing a sparse network to take advantage of this technology could further improve performance on AMLNN.